Examiner only

(c) The table below shows the area of fully organic land used to produce crops from 2013 to 2016.

	Area of fully organic land use, 2013 to 2016 (thousand hectares)			
	2013	**2014**	**2015**	**2016**
Cereals	43.7	42.2	39.6	38.4
Other crops	7.6	7.3	6.9	7.3
Fruit & nuts	2.1	2.1	1.9	1.9
Vegetables	11.3	9.4	10.4	10.2
Total	**64.7**	**61.0**	**58.8**	

Calculate the change in area of fully organic land used to produce these crops from 2013 to 2016. [2]

Change in area = ... thousand hectares

(d) The table below shows how organic livestock numbers have changed from 2013 to 2016.

	Organic livestock numbers 2013 to 2016 (thousand)			
	2013	**2014**	**2015**	**2016**
Cattle	283.3	304.1	291.5	296.4
Sheep	999.2	954.9	844.6	840.8
Pigs	30.2	28.3	30.0	31.5
Poultry	2487.6	2398.7	2560.2	2821.2

(i) State **one** ethical argument against the intensive farming of livestock. [1]

....

(ii) Explain whether the data in the table supports that this ethical argument has changed livestock farming practices. [2]

....

....

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(b) **Figure 1** shows the number and rate of reported cases of food poisoning in Wales between 1992 and 2016.

Figure 1

Year	Total number of reported cases	Rate (number per 100 000 population)
1992	3 590	124.5
1998	5 946	205.0
2000	4 716	162.2
2006	4 301	145.4
2010	4 980	165.6
2014	4 516	146.1
2016	4 504	145.7

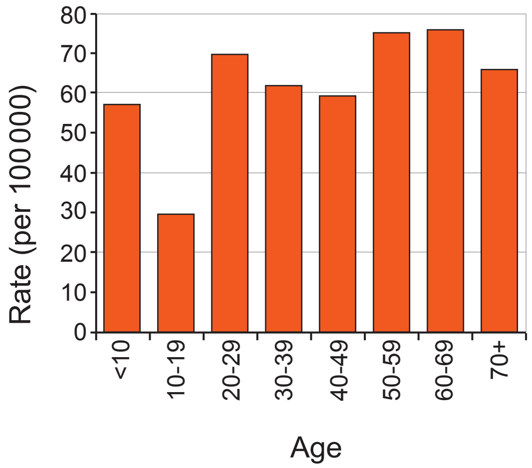
The percentage of reported cases of food poisoning during each quarter of a year is shown in **Figure 2**.

Figure 2

Quarter	% of total for year		
	2012	2014	2016
Jan-Mar	18	17	19
Apr-Jun	26	26	25
Jul-Sep	32	32	33
Oct-Dec	24	25	23

Figure 3 shows the rate of food poisoning amongst females by age group in 2016.

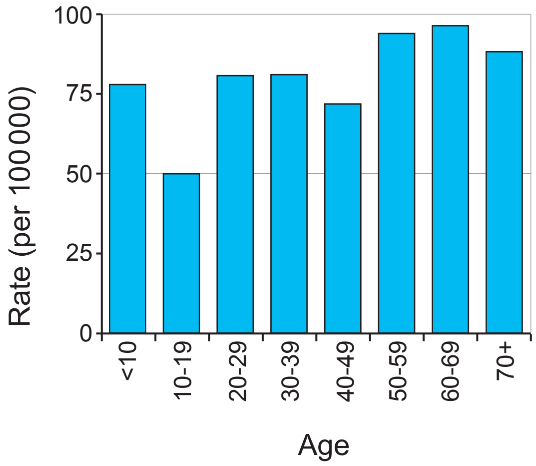
Figure 3



Examiner only

Figure 4 shows the rate of food poisoning amongst males by age group in 2016.

Figure 4



Use the data in **Figures 1 to 4** to answer the following questions.

(i) Since 1998, egg-laying hens have been vaccinated against a species of *Salmonella*. Explain how this has affected the number of reported cases of food poisoning. [2]

....

....

....

(ii) Explain the variation in the percentage of reported cases of food poisoning in each of the quarters in a year. [3]

....

....

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(iii) It is suggested that the rate of food poisoning does not depend on gender or age. Explain whether this suggestion is confirmed by the data. [2]

....

....

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THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number